Lecture 5 Activity- Relations and Functions

Question 1

Student= {Amal, Ben, Cara, Dan}

Subject= {Maths, IT, Science}

Grade= {A,B,C}

grades: Student <-> Grade

grades = {Amal |-> A, Ben |-> B, Cara |-> A, Dan |-> C}

enrolment: Student <-> Subject

enrolment = {Amal |-> IT, Amal |-> Maths, Ben |-> Science, Cara |-> IT, Dan |-> Maths}

Determine the following

1. Is grade a relation or a function? If it's a function is it a partial function or total function? Is it injective and/ or surjective?
2. dom(grades)
3. (Student * Grade)- grades
4. enrolment~ ; grade
5. enrolment [{Amal}]

Question 2

Vehicle = {Car, Bus, Bike, Boat, Plane}

Fuel= {Petrol, Diesel, Electric, None}

Capacity= {One, Few, Many}

fuelType: Vehicle <-> Fuel

fuelType = {Car |-> Petrol, Bus |-> Diesel, Bike |-> Electric, Boat |-> Diesel, Plane |-> Petrol }

capacity: Vehicle <-> Capacity

capacity = {Car |-> Few, Car |-> Many, Bus |-> Many, Bike |-> One, Boat |-> Few, Plane |-> Many}

Determine the following

1. Is fuelType a relation or a function. If it's a function is it a partial function or total function?
2. Vehicle * Capacity
3. ran(fuelType)
4. capacity~ ; fuelType
5. capacity[{Car, Boat}]

Question 3

Animal = {Cat, Dog, Fish, Bird}

Sound = {Meow, Bark, Chirp}

animalSound = {Cat |-> Meow, Dog |-> Bark}

Determine the following

1. Is animalSound a relation or a function? Is it a partial function or a total function? Is it injective and or surjective.
2. {Cat} <<| animalSound
3. (Animal * Sound) – animalSound
4. dom(animalSound)
5. animalSound[{Cat, Fish}]

Coding Question

Create a 2 Enumerated sets named Cars and Fuel. Create a total surjective function named fuelType that maps between Car and Fuel. Initialize with values

Implement the following operations

1. getFuelType(car)- returns fuel type of a specific car
2. getCar(fuel)- returns all car brands that use the given fuel type.
3. getAllCars- returns the set of all car brands
4. getAllFuelType- returns fuel types used by cars
5. isElectric- Check if car is electric and return Yes if so else return No